

LLM-Nmap

Network Scanning Using Large
Language Models



LLM-Nmap

Network Security – University of Pisa

Project: *LLM-Nmap, a network scanning using LLM*

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Project Overview and Goal

Use a **Large Language Model (LLM)** to drive **Nmap** using natural language:

- *"Scan this host and show me which services are running"*

The framework should:

- Take a **text description** of the desired analysis
- Let the **LLM choose and run Nmap** (via llm-tools-nmap.py)
- **Collect** the scan results and **present** them to the user

Deliverables:

- A **technical report** describing how to set up the full framework and use it
- A **demo** showing the complete pipeline working end to end



Scan this network

Background: Nmap

- **Nmap** = widely used **network scanner** for:

- Host discovery
- Port scanning
- Service and version detection

- **Limitation:**

- User must **remember the correct syntax and flags** for each task

```
nmap -sV -p- -A 10.0.2.X
```

```
Host is up (0.0015s latency).
```

```
Not shown: 995 closed ports.
```

PORT	STATE	SERVICE	VERSION
22	open	ssh	OpenSSH_6.2p1 Ubuntu
25	open	smtp	Postfix smtpd
80	open	http	Apache/2.4.18 (Ubuntu)
443	open	https	Apache/2.4.18 (Ubuntu)



Background: LLM and LLM-Nmap

LLMs (e.g. Gemini):

- Understand natural language prompts
- Can choose tools / functions to call with arguments

llm CLI:

- Command-line interface to multiple LLM providers
- Supports plugins / “functions” to integrate external tools

llm-tools-nmap.py:

- Existing plugin that exposes Nmap operations as functions

How it runs:

- Input to the LLM: user prompt + available function definitions
- LLM chooses the right Nmap scan and returns the result/output

System Architecture and Environment

Kali VM:

- Runs llm + Gemini + Nmap
- Loads llm-tools-nmap.py (LLM-Nmap controller)

Ubuntu Server VM:

- “Normal” target
- We open/close SSH on port 22 for tests

BeeBox VM:

- Deliberately vulnerable target
- Many open ports and services

Mac laptop:

- Physical target host on the same network

All VMs attached to a **VirtualBox NAT Network** (e.g. 10.0.2.0/24).



Network Setup and Connectivity

Each VM obtains an IP address in the NAT network (e.g. 10.0.2.x).

Example:

From **Kali VM** we check connectivity:

- ping 10.0.2.10 – Ubuntu
- ping 10.0.2.20 – BeeBox

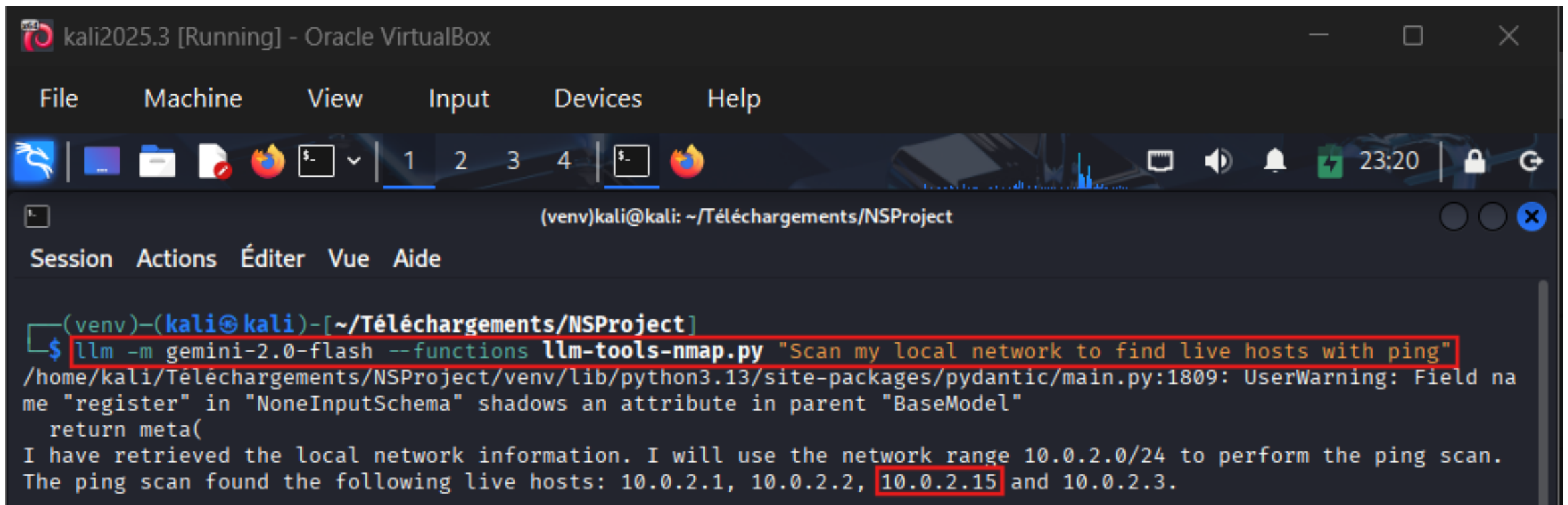
Successful pings confirm:

- All machines are on the same NAT network
- Targets are **reachable** for Nmap scans



First LLM–Nmap Test: Network Scan

- Prompt from Kali:
“Scan my local network to find live hosts with ping”



The screenshot shows a Kali Linux terminal window titled "kali2025.3 [Running] - Oracle VirtualBox". The terminal is running a command to use an LLM for a network scan. The command is: `llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Scan my local network to find live hosts with ping"`. The output shows a warning from Pydantic and then a response from the LLM: "I have retrieved the local network information. I will use the network range 10.0.2.0/24 to perform the ping scan. The ping scan found the following live hosts: 10.0.2.1, 10.0.2.2, 10.0.2.15 and 10.0.2.3." The IP address 10.0.2.15 is highlighted in red in the original image.

```
(venv)kali@kali: ~/Téléchargements/NSProject
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Scan my local network to find live hosts with ping"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
I have retrieved the local network information. I will use the network range 10.0.2.0/24 to perform the ping scan.
The ping scan found the following live hosts: 10.0.2.1, 10.0.2.2, 10.0.2.15 and 10.0.2.3.
```


First LLM–Nmap Test: Network Scan

1

LLM sends the prompt + functions from llm-tools-nmap.py to Gemini.

2

Gemini chooses the **host-discovery scan**.

3

Nmap runs from Kali on the NAT subnet.

First LLM–Nmap Test: Results

Live hosts (including the **Ubuntu Server** VM) appear in the output.

LLM–Nmap can translate a **natural-language request** into a valid Nmap scan.

Scenario 1 – Kali VM Scanning Ubuntu



Kali → Ubuntu
Server VM

Target: Ubuntu
Server VM at
10.0.2.15

Scenario 1 – Step 1



LLM-Nmap

Before opening SSH port:

- **Prompt:** “Make a quick scan of 10.0.2.15”
- **LLM-Nmap result:** No open ports found.

```
kali2025.3 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
(venv)kali@kali: ~/Téléchargements/NSProject
Session Actions Éditer Vue Aide
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.15"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
The quick scan of 10.0.2.15 shows that all 100 scanned ports are in ignored states. The MAC address is 08:00:27:B1:B8:74, which belongs to PCS Systemtechnik/Oracle VirtualBox virtual NIC. The host is up.
```

```
UbuntuServerNSProject [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@UbuntuServerNSProject:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:feb1:b874 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b1:b8:74 txqueuelen 1000 (Ethernet)
    RX packets 241 bytes 246873 (246.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 130 bytes 13925 (13.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Scenario 1 – Step 2



After opening SSH port

- Same prompt: “Make a quick scan of 10.0.2.15”
- LLM-Nmap: Detects port 22/tcp open

```
kali2025.3 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
(venv)kali@kali: ~/Téléchargements/NSProject
Session Actions Éditer Vue Aide
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.15"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field na
me "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
The quick scan of 10.0.2.15 shows that all 100 scanned ports are in ignored states. The MAC address is 08:00:27:B1:
B8:74, which belongs to PCS Systemtechnik/Oracle VirtualBox virtual NIC. The host is up.

(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.15"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field na
me "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
The quick scan of 10.0.2.15 shows that port 22 is open, indicating an SSH service is likely running. The MAC address
s is 08:00:27:B1:B8:74, which belongs to PCS Systemtechnik/Oracle VirtualBox.
```


Scenario 1 – Step 3



Manual Nmap:

- **Command from Kali:** `nmap -sV 10.0.2.15`
- **Manual Nmap output:** Shows **port 22/tcp open (SSH service)**

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.15"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
The quick scan of 10.0.2.15 shows that port 22 is open, indicating an SSH service is likely running. The MAC address is 08:00:27:B1:B8:74, which belongs to PCS Systemtechnik/Oracle VirtualBox.

(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ nmap 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-05 00:02 CET
Nmap scan report for 10.0.2.15
Host is up (0.00051s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:B1:B8:74 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds
```

Scenario 1 – LLM–Nmap vs Manual Nmap



Comparison:

LLM–Nmap and manual Nmap report **the same open port and service.**

Conclusion:

The framework correctly:

- Configures and runs Nmap via the LLM
- Returns accurate results to the user

Scenario 2 – Kali VM Scanning BeeBox



Kali → BeeBox

Target: BeeBox
VM at 10.0.2.7

Scenario 2 – Step 1



LLM-Nmap

- **Prompt:** "Make a quick scan of 10.0.2.7"
- The scan found 13 different open ports.

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.5-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.7"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field na
me "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
I have completed a quick scan of 10.0.2.7. Here are the open ports and their services:

* **21/tcp:** ftp
* **22/tcp:** ssh
* **25/tcp:** smtp
* **80/tcp:** http
* **139/tcp:** netbios-ssn
* **443/tcp:** https
* **445/tcp:** microsoft-ds
* **513/tcp:** login
* **514/tcp:** shell
* **3306/tcp:** mysql
* **6001/tcp:** X11:1
* **8080/tcp:** http-proxy
* **8443/tcp:** https-alt

The MAC Address is 08:00:27:77:53:57 (PCS Systemtechnik/Oracle VirtualBox virtual NIC).
```

13 ports

Scenario 2 – Step 2



Manual nmap

- Command from Kali nmap 10.0.2.7

17 ports

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ nmap 10.0.2.7
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-13 18:28 CET
Nmap scan report for 10.0.2.7
Host is up (0.00076s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
666/tcp   open  doom
3306/tcp  open  mysql
5901/tcp  open  vnc-1
6001/tcp  open  X11:1
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
9080/tcp  open  glrpc
MAC Address: 08:00:27:77:53:57 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.46 seconds
```


Scenario 2 – LLM–Nmap vs Manual Nmap (1)



Comparison:

- Nmap reveals that **17 ports** were open
- LLM-nmap found only **13 ports**

Conclusion:

With the basic instructions, llm-nmap didn't succeed in finding all open ports (4 are missing compared to Manual Nmap).

Scenario 2 – Step 3



Challenging llm-nmap

- Changing "quick" to "deep" and see the results

11 ports

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.5-flash --functions llm-tools-nmap.py "Make a deep scan of 10.0.2.7"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(I have performed a deep scan on 10.0.2.7. Here are the findings:

**Host Information:**
* **IP Address:** 10.0.2.7
* **MAC Address:** 08:00:27:77:53:57 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
* **Device type:** general purpose
* **Operating System:** Linux 2.6.X (specifically detected as Linux 2.6.13 - 2.6.32, Linux 2.6.18 - 2.6.24)
* **Hostname:** bee-box
* **Network Distance:** 1 hop

**Open Ports and Services:**
* **21/tcp (ftp):** ProFTPD 1.3.1
  * **Details:** Anonymous FTP login is allowed, and several PDF files are listed (e.g., Iron_Man.pdf, Terminat
or_Salvation.pdf, bWAPP_intro.pdf).
* **22/tcp (ssh):** OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
  * **Host Keys:** DSA (1024 bit), RSA (2048 bit)
* **25/tcp (smtp):** Postfix smtpd
  * **SSL/TLS:** SSLv2 supported with various ciphers. Certificate subject: commonName=ubuntu. Not valid after
2013-04-27.
* **SMTP Commands:** PIPELINING, SIZE 1024000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN.
* **80/tcp (http):** Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Patch m
od_ssl/2.2.8 OpenSSL/0.9.8g)
  * **Details:** Potentially risky methods: TRACE. Site doesn't have a title.
* **139/tcp (netbios-ssn):** Samba smbd 3.0.28a (workgroup: ITSECGAMES)
* **443/tcp (ssl/http):** Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Pa
tch mod_ssl/2.2.8 OpenSSL/0.9.8g)
  * **SSL/TLS:** SSLv2 supported with various ciphers. Certificate subject: commonName=bee-box.bwapp.local. Not
valid after 2018-04-13.
  * **Details:** Potentially risky methods: TRACE. Site doesn't have a title.
* **445/tcp (netbios-ssn):** Samba smbd 3.0.28a (workgroup: ITSECGAMES)
* **512/tcp (exec):** netkit-rsh rexecd
* **513/tcp (login?):** Service detected, but version unknown.
* **514/tcp (shell?):** Service detected, but version unknown.
* **666/tcp (doom?):**
```

6 ports

```
* **Details:** Potentially risky methods: TRACE. Site doesn't have a title.
* **445/tcp (netbios-ssn):** Samba smbd 3.0.28a (workgroup: ITSECGAMES)
* **512/tcp (exec):** netkit-rsh rexecd
* **513/tcp (login?):** Service detected, but version unknown.
* **514/tcp (shell?):** Service detected, but version unknown.
* **666/tcp (doom?):**
  * **Details:** Returns "*** bWAPP Movie Service *** Matching movies: 0"
* **3306/tcp (mysql):** MySQL 5.0.96-0ubuntu3
  * **Details:** Protocol 10, Thread ID 39. Supports Compression, ConnectWithDatabase, Speaks41ProtocolNew, Sup
port41Auth, LongColumnFlag, SupportsTransactions. Status: Autocommit.
* **5901/tcp (vnc):** VNC (protocol 3.8)
  * **Security Types:** VNC Authentication (2)
* **6001/tcp (X11):** (access denied)
* **8080/tcp (http):** nginx 1.4.0
  * **Details:** Appears to be an open proxy. Site doesn't have a title.
* **8443/tcp (ssl/http):** nginx 1.4.0
  * **SSL/TLS:** Certificate subject: commonName=bee-box.bwapp.local. Not valid after 2018-04-13.
  * **TLS Next Protocol Negotiation:** http/1.1
  * **Details:** Site doesn't have a title.
* **9080/tcp (http):** lighttpd 1.4.19
  * **Details:** Site doesn't have a title.

**SMB Security Mode:**
* 'account_used': guest
* 'authentication_level': user
* 'challenge_response': supported
* 'message_signing': disabled (dangerous, but default)

**SMB OS Discovery:**
* 'OS': Unix (Samba 3.0.28a)
* 'Computer name': bee-box
* 'FQDN': bee-box
* 'System time': 2026-01-13T06:01:04+01:00

**Clock Skew:**
* 'mean': -12h43m52s
* 'deviation': 26m49s
* 'median': -12h31m52s

The scan took 193.24 seconds.
```

Scenario 2 – LLM–Nmap vs Manual Nmap (2)



Conclusion:

- Request took longer to complete
- Provided better results (*similar to using the -A option with Manual Nmap*)
- All ports that the "basic" Nmap request found were found

Scenario 3 – Kali VM Scanning a Mac Laptop



Kali → Mac
Laptop

Target: Mac
laptop at
10.184.43.104

Scenario 3 – Kali VM Scanning a Mac Laptop



- Identified **IP address 10.184.43.104**
- Checked which **ports/services** are open

```
tommy@Tommys-MacBook-Air ~ % netstat -an | grep LISTEN
tcp6      0      0  *.64322          *.*          LISTEN
tcp4      0      0  *.64322          *.*          LISTEN
tcp6      0      0  *.5000           *.*          LISTEN
tcp4      0      0  *.5000           *.*          LISTEN
tcp6      0      0  *.7000           *.*          LISTEN
tcp4      0      0  *.7000           *.*          LISTEN
tcp46     0      0  *.53378          *.*          LISTEN
tommy@Tommys-MacBook-Air ~ %
```

Scenario 3 – Step 1



LLM–Nmap:

- **Prompt:** *“Make a quick scan on 10.184.43.104”*

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan on 10.184.43.104"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
The quick scan on 10.184.43.104 shows that port 5000 is open and running the service 'upnp'.
```

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Make a quick scan on 10.184.43.104 up to port 10000"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
The scan of 10.184.43.104 on ports 1-10000 shows that ports 5000 and 7000 are open.
```


Scenario 3 – Step 2



Manual Nmap scan

- **Command from Kali:** `nmap -A <mac-ip>`

```
(kali㉿kali)-[~/Téléchargements/NSProject]
$ nmap -A 10.184.43.104
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-05 15:49 CET
Nmap scan report for 10.184.43.104
Host is up (0.0018s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5000/tcp  open  rtsp
|_rtsp-methods: ERROR: Script execution failed (use -d to debug)
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 403 Forbidden
|     Content-Length: 0
|     Server: AirTunes/920.10.1
|     X-Apple-ProcessingTime: 1
|     X-Apple-RequestReceivedTimestamp: 96100899
|   GetRequest:
|     HTTP/1.1 403 Forbidden
|     Content-Length: 0
|     Server: AirTunes/920.10.1
|     X-Apple-ProcessingTime: 2
|     X-Apple-RequestReceivedTimestamp: 96095768
```

```
HTTPOptions:
|   HTTP/1.1 403 Forbidden
|   Content-Length: 0
|   Server: AirTunes/920.10.1
|   X-Apple-ProcessingTime: 3
|   X-Apple-RequestReceivedTimestamp: 96100866
| RTSPRequest:
|   RTSP/1.0 403 Forbidden
|   Content-Length: 0
|   Server: AirTunes/920.10.1
|   X-Apple-ProcessingTime: 3
|   X-Apple-RequestReceivedTimestamp: 96095852
| SIPOptions:
|   RTSP/1.0 403 Forbidden
|   Content-Length: 0
|   Server: AirTunes/920.10.1
|   CSeq: 42 OPTIONS
|   X-Apple-ProcessingTime: 2
|   X-Apple-RequestReceivedTimestamp: 96100930
7000/tcp  open  rtsp
|_rtsp-methods: ERROR: Script execution failed (use -d to debug)
| fingerprint-strings:
```

Scenario 3 – Kali VM Scanning a Mac Laptop



Comparison

For the Mac host, LLM–Nmap and manual Nmap:

- Report **the same set of open ports**.
- Report **consistent services** on those ports.

Conclusion

The difference is mainly in presentation:

- LLM–Nmap wraps the raw Nmap output in a more “human” explanation.
- The framework works not only in virtual network, but also in a real network.

Limitations

Dependence on the LLM:

- Quality of the scan depends on how the LLM interprets the prompt.
- Not specified prompt can lead to random answers
- Inconsistent for the same request

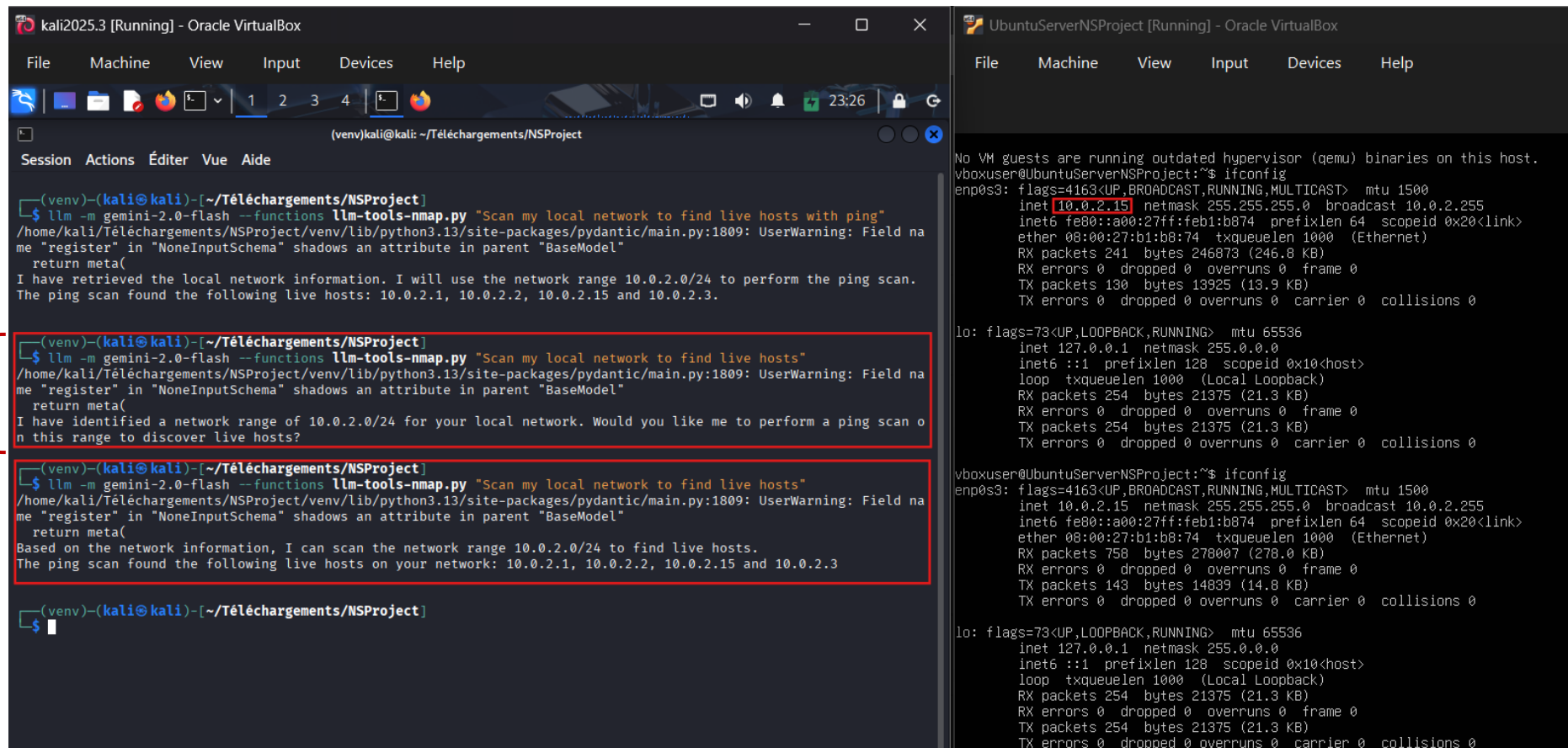
Performance / API limits:

- Online LLM (Gemini) introduces **latency**.
- API keys and **rate limits** may apply
- We couldn't use other models like (ChatGPT or Claude)

Limitations – No answer

No answer

Sometimes llm-nmap doesn't respond to a request and instead replies with a question.



```
kali2025.3 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
(venv)kali@kali: ~/Téléchargements/NSProject
Session Actions Éditer Vue Aide
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Scan my local network to find live hosts with ping"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
I have retrieved the local network information. I will use the network range 10.0.2.0/24 to perform the ping scan.
The ping scan found the following live hosts: 10.0.2.1, 10.0.2.2, 10.0.2.15 and 10.0.2.3.

(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Scan my local network to find live hosts"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
I have identified a network range of 10.0.2.0/24 for your local network. Would you like me to perform a ping scan on this range to discover live hosts?

(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.0-flash --functions llm-tools-nmap.py "Scan my local network to find live hosts"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
Based on the network information, I can scan the network range 10.0.2.0/24 to find live hosts.
The ping scan found the following live hosts on your network: 10.0.2.1, 10.0.2.2, 10.0.2.15 and 10.0.2.3

(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$

UbuntuServerNSProject [Running] - Oracle VirtualBox
File Machine View Input Devices Help
No VM guests are running outdated hypervisor (qemu) binaries on this host.
vboxuser@UbuntuServerNSProject:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:feb1:b874 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b1:b8:74 txqueuelen 1000 (Ethernet)
    RX packets 241 bytes 246873 (246.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 130 bytes 13925 (13.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 254 bytes 21375 (21.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 254 bytes 21375 (21.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vboxuser@UbuntuServerNSProject:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:feb1:b874 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b1:b8:74 txqueuelen 1000 (Ethernet)
    RX packets 758 bytes 278007 (278.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 143 bytes 14839 (14.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 254 bytes 21375 (21.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 254 bytes 21375 (21.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Limitations – Consistency of results

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.5-flash --functions llm-tools-nmap.py "Make a deep scan of 10.0.2.7"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
I have performed a deep scan on 10.0.2.7. Here are the findings:

**Host Information:**
* **IP Address:** 10.0.2.7
* **MAC Address:** 08:00:27:77:53:57 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
* **Device type:** general purpose
* **Operating System:** Linux 2.6.X (specifically detected as Linux 2.6.13 - 2.6.32, Linux 2.6.18 - 2.6.24)
* **Hostname:** bee-box
* **Network Distance:** 1 hop

**Open Ports and Services:**
* **21/tcp (ftp):** ProFTPD 1.3.1
  * **Details:** Anonymous FTP login is allowed, and several PDF files are listed (e.g., Iron_Man.pdf, Terminator_Salvation.pdf, bWAPP_intro.pdf).
* **22/tcp (ssh):** OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
  * **Host Keys:** DSA (1024 bit), RSA (2048 bit)
* **25/tcp (smtp):** Postfix smtpd
  * **SSL/TLS:** SSLV2 supported with various ciphers. Certificate subject: commonName=ubuntu. Not valid after 2013-04-27.
* **SMTP Commands:** PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN.
* **80/tcp (http):** Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Patch mod_ssl/2.2.8 OpenSSL/0.9.8g)
  * **Details:** Potentially risky methods: TRACE. Site doesn't have a title.
* **139/tcp (netbios-ssn):** Samba smbd 3.X - 4.X (workgroup: ITSECgames)
* **443/tcp (ssl/http):** Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Patch mod_ssl/2.2.8 OpenSSL/0.9.8g)
  * **SSL/TLS:** SSLV2 supported with various ciphers. Certificate subject: commonName=bee-box.bwapp.local. Not valid after 2018-04-13.
  * **Details:** Potentially risky methods: TRACE. Site doesn't have a title.
* **445/tcp (netbios-ssn):** Samba smbd 3.0.28a (workgroup: ITSECgames)
* **512/tcp (exec):** netkit-rsh rexecd
* **513/tcp (login):** Service detected, but version unknown.
* **514/tcp (shell):** Service detected, but version unknown.
* **666/tcp (doom?):**
  * **Details:** Returns "*** bWAPP Movie Service ** Matching movies: 0"
* **3306/tcp (mysql):** MySQL 5.0.96-0ubuntu3
  * **Details:** Protocol 10, Thread ID 39. Supports Compression, ConnectWithDatabase, Speaks41ProtocolNew, Super41Auth, LongColumnFlag, SupportsTransactions. Status: Autocommit.
* **5901/tcp (vnc):** VNC (protocol 3.8)
  * **Security Types:** VNC Authentication (2)
* **6001/tcp (X11):** (access denied)
* **8080/tcp (http):** nginx 1.4.0
  * **Details:** Appears to be an open proxy. Site doesn't have a title.
* **8443/tcp (ssl/http):** nginx 1.4.0
  * **SSL/TLS:** Certificate subject: commonName=bee-box.bwapp.local. Not valid after 2018-04-13.
  * **TLS Next Protocol Negotiation:** http/1.1
  * **Details:** Site doesn't have a title.
* **9080/tcp (http):** lighttpd 1.4.19
  * **Details:** Site doesn't have a title.

**SMB Security Mode:**
* 'account_used': guest
* 'authentication_level': user
* 'challenge_response': supported
* 'message_signing': disabled (dangerous, but default)

**SMB OS Discovery:**
* OS: Unix (Samba 3.0.28a)
* Computer name: bee-box
* FQDN: bee-box
* System time: 2026-01-13T06:01:04+01:00

**Clock Skew:**
* 'mean': -12h43m52s
* 'deviation': 26m49s
* 'median': -12h31m52s

The scan took 193.24 seconds.
```

Different depth of responses and results:

- Size
- Running time
- Details

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.5-flash --functions llm-tools-nmap.py "Make a deep scan of 10.0.2.7"
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta()
The deep scan of 10.0.2.7 revealed the following open ports and services:

* **21/tcp:** FTP - ProFTPD 1.3.1 (Anonymous FTP login allowed, several PDF files listed)
* **22/tcp:** SSH - OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
* **25/tcp:** SMTP - Postfix smtpd (SSLV2 supported, certificate details provided)
* **80/tcp:** HTTP - Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Patch mod_ssl/2.2.8 OpenSSL/0.9.8g)
* **139/tcp:** NetBIOS-ssn - Samba smbd 3.X - 4.X
* **443/tcp:** SSL/HTTP - Apache httpd 2.2.8 ((Ubuntu) DAV/2 mod_fastcgi/2.4.6 PHP/5.2.4-2ubuntu5 with Suhosin-Patch mod_ssl/2.2.8 OpenSSL/0.9.8g) (SSLV2 supported, certificate details provided)
* **445/tcp:** NetBIOS-ssn - Samba smbd 3.0.28a
* **512/tcp:** Exec - netkit-rsh rexecd
* **513/tcp:** Login?
* **514/tcp:** Shell?
* **666/tcp:** Doom? (unrecognized service, but returned "*** bWAPP Movie Service **")
* **3306/tcp:** MySQL - MySQL 5.0.96-0ubuntu3
* **5901/tcp:** VNC (protocol 3.3; Locked out)
* **6001/tcp:** X11 (access denied)
* **8080/tcp:** HTTP - nginx 1.4.0 (Potentially acting as an open proxy)
* **8443/tcp:** SSL/HTTP - nginx 1.4.0
* **9080/tcp:** HTTP - lighttpd 1.4.19

The operating system detected is **Linux 2.6.X**.
The NetBIOS name is BEE-BOX with the workgroup ITSECgames.
Anonymous FTP login is allowed on port 21, and several PDF files are accessible.
SSLV2 is supported on ports 25 (SMTP) and 443 (HTTPS), which is considered insecure.
The VNC service on port 5901 is locked out.
The HTTP service on port 8080 running nginx 1.4.0 is reported as a potential open proxy.
```

Conclusion

Implemented a framework where:

- User describes an analysis in **natural language**.
- LLM (Gemini) + llm-tools-nmap.py run the appropriate Nmap scan.
- Results are returned and compared with manual Nmap.
- Validated the framework in **three scenarios**

Overall, the project confirms that LLMs can make network scanning more accessible by bridging between natural language and powerful but complex tools like Nmap.

Conclusion

- We can also use llm-nmap to ask it to perform tasks that Nmap **can't do, such as generating reports.**

```
(venv)-(kali@kali)-[~/Téléchargements/NSProject]
$ llm -m gemini-2.5-flash --functions llm-tools-nmap.py "Make a quick scan of 10.0.2.7 and make me a report of the result. Highlight network vulnerabilities."
/home/kali/Téléchargements/NSProject/venv/lib/python3.13/site-packages/pydantic/main.py:1809: UserWarning: Field name "register" in "NoneInputSchema" shadows an attribute in parent "BaseModel"
  return meta(
Here's a report based on the quick Nmap scan of 10.0.2.7:

**Nmap Quick Scan Report for 10.0.2.7**

**Scan Date:** January 13, 2026
**Target IP:** 10.0.2.7
**Host Status:** Host is up (0.0012s latency).
**MAC Address:** 08:00:27:77:53:57 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

**Open Ports and Services:**

The following ports were found to be open on the target system:

* **21/tcp:** ftp (File Transfer Protocol)
* **22/tcp:** ssh (Secure Shell)
* **25/tcp:** smtp (Simple Mail Transfer Protocol)
* **80/tcp:** http (Hypertext Transfer Protocol)
* **139/tcp:** netbios-ssn (NetBIOS Session Service)
* **443/tcp:** https (Secure HTTP)
* **445/tcp:** microsoft-ds (Microsoft Directory Services - SMB)
* **513/tcp:** login (rlogin)
* **514/tcp:** shell (rsh)
* **3306/tcp:** mysql (MySQL Database)
* **6001/tcp:** X11:1 (X Window System)
* **8080/tcp:** http-proxy (Alternative HTTP)
* **8443/tcp:** https-alt (Alternative HTTPS)
```


Conclusion

- We can also use llm-nmap to ask it to perform tasks that Nmap can't do, such as generating reports.

****Identified Network Vulnerabilities and Risks:****

Based on the open ports and services, **several potential network vulnerabilities and risks can be highlighted:**

- **21/tcp (FTP):****
 - **Vulnerability:**** FTP is inherently insecure as it often transmits credentials and data in plaintext. Anonymous FTP access, if enabled, can lead to unauthorized file access or upload of malicious content. Outdated FTP server software may also have known exploits.
 - **Risk:**** Data breaches, unauthorized file modification, system compromise.
- **25/tcp (SMTP):****
 - **Vulnerability:**** If not properly secured, an SMTP server can be used for spam relaying, email spoofing, or information gathering through user enumeration.
 - **Risk:**** Reputation damage, phishing attacks, disclosure of user accounts.
- **80/tcp (HTTP), 8080/tcp (HTTP-Proxy), 443/tcp (HTTPS), 8443/tcp (HTTPS-alt):****
 - **Vulnerability:**** Web servers are frequent targets. Common vulnerabilities include SQL Injection, Cross-Site Scripting (XSS), Directory Traversal, misconfigurations, and exploits in web applications or server software (e.g., Apache, Nginx). For HTTPS, weak SSL/TLS configurations, outdated protocols, or expired/self-signed certificates can reduce security.
 - **Risk:**** Data breaches, website defacement, session hijacking, denial of service, remote code execution.
- **139/tcp (netbios-ssn) & 445/tcp (microsoft-ds - SMB):****
 - **Vulnerability:**** These services are commonly associated with Windows file sharing (SMB/CIFS). They have been historically vulnerable to critical exploits like EternalBlue (used in WannaCry), brute-force attacks on shared resources, and information disclosure (e.g., listing shares, users, system information). Misconfigured shares can lead to unauthorized access to sensitive files.
 - **Risk:**** Remote code execution, unauthorized file access, data theft, propagation of malware.
- **513/tcp (login - rlogin) & 514/tcp (shell - rsh):****
 - **Vulnerability:**** These are legacy Unix 'r-services' that are highly insecure. They often rely on host-based authentication (`.rhosts` files) which is easily spoofed, and they transmit credentials and data in plaintext. Their continued presence is a significant security risk.
 - **Risk:**** Unauthorized remote command execution, complete system compromise, credential theft. ****It is strongly recommended to disable these services.****
- **3306/tcp (MySQL):****
 - **Vulnerability:**** Database servers are critical assets. Weak passwords, default credentials, unpatched MySQL server vulnerabilities, or improper access controls can lead to unauthorized database access. Applications interacting with the database may also be vulnerable to SQL injection.
 - **Risk:**** Data breaches, data manipulation, denial of service, complete system compromise if the database runs with elevated privileges.
- **6001/tcp (X11):****
 - **Vulnerability:**** If X11 is configured to allow remote connections without proper authentication (e.g., `xhost +`), it can lead to unauthorized access to the graphical desktop environment. Attackers could potentially inject key strokes, capture screenshots, or execute commands.
 - **Risk:**** Information disclosure, remote code execution, full system compromise.

Conclusion

- We obtained a complete cybersecurity report including the open ports found, potential risks, and recommendations to improve the security of our network.

****General Recommendations:****

```
* **Disable Unnecessary Services:** Review all open ports and disable any services that are not strictly required. The presence of `rlogin` (513/tcp) and `rsh` (514/tcp) is particularly concerning and these should be disabled immediately.
* **Patch and Update:** Ensure all operating systems, web servers, database servers, and other software are regularly patched and updated to the latest stable versions to protect against known vulnerabilities.
* **Strong Authentication:** Implement strong, unique passwords for all services (SSH, FTP, MySQL, etc.) and consider using key-based authentication for SSH.
* **Firewall Rules:** Configure a firewall to restrict access to services from untrusted networks. Only allow necessary traffic from authorized sources.
* **Secure Configurations:**
  * For FTP, consider using SFTP or FTPS for encrypted data transfer.
  * For SSH, disable root login, enforce strong ciphers, and limit user access.
  * For web servers (HTTP/HTTPS), implement web application firewalls (WAFs), regularly scan for web vulnerabilities, and ensure secure SSL/TLS configurations.
  * For SMB (139/tcp, 445/tcp), ensure the latest security updates are applied and enforce strong authentication and authorization on shared resources.
  * For MySQL, restrict remote access to trusted IPs, use strong credentials, and limit user privileges.
  * For X11, ensure it is not exposed to the internet, and use X11 forwarding over SSH for secure remote access.
* **Regular Audits:** Conduct regular security audits and vulnerability scans to identify and address new risks.
```

This quick scan provides a snapshot of potential entry points. Further, more in-depth scans (e.g., service version detection, vulnerability scripts) would be needed to confirm specific vulnerabilities and their severity.

Future Work

1

Parse Nmap XML / structured output to generate **tables or dashboards**.

2

Compare multiple LLMs (Gemini vs GPT vs Claude) using the same plugin.

3

Integrate other security tools as additional LLM “functions”.



Thank You

Any questions?